The Open Source Firefighter

Design Philosophy — The Internal Rule Set

v1.0 · August 2026 · The rules for what goes into the box *Document ID:* ***OSF-0*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

These are not requirements for the equipment; those live in the specification. These are the rules the specification was written by — made explicit so that future contributors, and future versions of the authors, apply the same filter.

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1. Invent only at the coordination layer. Everything else is borrowed: valves from industry, radios from their ecosystems, routing from forty years of networking, fire science from the agencies, drying math from NFDRS. If a problem can be solved by citing someone else's solved problem, cite it. Novelty is a budget, and it is spent in exactly one place.

2. Novelty is spent only where the invention lives — everything the hands touch is familiar. The box resembles a fire alarm panel on purpose. Resemblance is inherited trust: inspectors approve what they recognize, trades already know the work, and society already runs the maintenance habits. The enclosure earns trust by familiarity so the invention inside never has to spend any.

3. Time-share the constraint; never pretend it away. A hose bib, a fuel can, a municipal main — the supply that exists is the design input. Sequencing is the same algorithm at every scale: the box time-shares a spigot, the street time-shares a main, the domain time-shares a valley. Honest hydraulics first; the physics is also the fallback delivery mechanism.

4. Enhancement, never dependency. Every capability beyond the core — backhaul, weather feeds, satellite SOS, incident integration — makes the system better when present and changes nothing when absent. The system operates when isolated. This rule is absolute and unweakenable, and it decides most design questions before they are asked.

5. Safety is structural, not promised. A protocol with no string type cannot smuggle payloads. Data with no message category cannot be transported. A policy can be revised; an architecture has to be rebuilt. Where a boundary matters — surveillance, mission creep, enclosure — build it into what the system *can* do, not what it agrees to do.

6. Operators prevail. Automation manages resources; it never overrules the person standing at the equipment. Where policy and operator conflict at the point of use, the operator wins and the system makes consequences visible. The machine informs; it does not refuse.

7. The filter and the floor converge. Design against the hardest user — the expert who will stand on it, overload it, and defeat any lockout. Design for the newest user — the volunteer meeting it for the first time, at night, in gloves. These converge: simple, verbal, tactile, forgiving, heavy-handled, nothing delicate, nothing that requires reading. A choice serving only one has failed both.

8. Commodity wear items, forever. Every part with a service life is replaceable by a stranger with a supply house account and no documentation beyond the label. Single-source dependency is non-compliance by definition. The 2046 unit and the 2026 unit interoperate because both are made of things that never stopped being made.

9. Wrong category on purpose. At every regulatory interface, the equipment deliberately belongs to the open regime, not the moated one: an industrial control panel, not a fire-alarm control unit; unlicensed spectrum, not coordinated channels; a wetting system, not a life-safety device. Look like the panel; certify like the sprinkler timer. The incumbent's moat only works on equipment that enters their category — so don't.

10. Know whose house it is. Dispatch owns the trunks. The COML owns the incident spectrum. The homeowner owns the override. The agency owns its radios and its imagery. The system integrates by respecting existing authority at every tier — leaving the porch light on rather than knocking on doors. Restraint, stated in writing, is its own credential.

11. Degrade to the baseline, which is zero. Every failure mode returns to the unprotected status quo the world already accepts. A dead node is a house as it was; a missed packet is answered by pressure physics; no fix means run conservative and wet. Strictly additive by architecture — which is why the system never needs permission to exist.

12. See the weapon-shape early; weld the safety on while the metal is hot. Every capable system has a second squint. Notice it at design time and answer it in architecture — enumerated messages, mission-bounded charters, operator supremacy, irrevocable openness — so the safeguard is welded, not bolted, and holds when the author is tired, gone, or done.

13. Comprehension is the adoption mechanism. The system shall be fully comprehensible to the people who build, install, inspect, and live with it — the homeowner at their depth, the electrician at the print, the comms tech at the stack. A system with a reachable floor is one people teach each other, improve, and defend as theirs. We were never designing for approval; we were designing for understanding, and approval is what understanding does on its own. Corollary: where an architectural boundary exists, give it a visual expression the relevant inspecting culture already reads (orange wire stays hot; no red jackets on our rail) — legibility is verification at zero cost.

14. Adopt the pattern, drop the vestige. When borrowing a proven frame — a compliance standard, a trade practice, another device's interface — take the principle and shed the parts that solved a different problem. The headphone jack solved the ATM's privacy problem; a device with no private audio does not grow that tail. Cite the frame, inherit the intent, leave the appendages. This is the discipline that keeps borrowing from becoming accretion.

15. If you cannot tell a coherent operational story about it, you have not finished designing it. A design is complete when a realistic scenario can be told end to end — every actor doing something plausible, every failure degrading sensibly, every gap visible — without hand-waving. The places the story resists being told are the places the design has an unwritten assumption. Narrative coherence is an audit: build the story deliberately to find the seams, and populate it with real reviewers in distinct roles where you can, because the decision-architecture is the thing under test, not any single component. The strongest doctrine is usually not invented but remembered — operational memory transcribed and generalized — because remembered doctrine has already survived contact with reality and therefore has no seams.

The pattern under the patterns: constraints are the design engine. The charter, the layer model, and this list exist so that most decisions answer themselves by subtraction — the complicated version is usually a rule violation wearing a feature costume, and it keeps getting simpler because deleting an imported assumption deletes hardware.